

RegDivergence-101: An LLM Benchmark for Cross-Jurisdiction Regulatory Contradiction Detection in Life Sciences

CHUCHU WU*, Carnegie Mellon University, USA

ZHIYIN ZHOU, Pratt Institute, USA

JINGZHUO HU, University of Pennsylvania, USA

LIANG YOU, University of Pittsburgh, USA

Pharmaceutical sponsors developing a drug for both the United States and the European Union must reconcile guidance issued independently by the FDA and the EMA. Where the two agencies require *substantively the same thing*, a sponsor can file once; where they *diverge*, a single trial design risks rejection in one region; where one agency is *silent* on a point the other regulates, the sponsor must infer obligations. Today this reconciliation is performed manually by regulatory-affairs experts. We introduce **cross-jurisdiction regulatory divergence detection**: given an FDA requirement and an EMA requirement on the same topic, classify their relationship as AGREE, DIVERGE, or SILENT. SILENT is inherently directional (SILENT_FDA vs. SILENT_EMA); we record direction per pair and report per-direction F1 alongside the collapsed label. We release **RegDivergence-101**, a 101-pair expert-grounded **pilot** evaluation benchmark (labels grounded in three peer-reviewed FDA/EMA comparison studies and primary FDA/EMA/ICH guidance text; dual-annotation inter-annotator $\kappa = 0.85$), and systematically characterise a four-method baseline hierarchy: lexical heuristic (0.511 macro-F1, 95% CI [0.411–0.605]), NLI cross-encoder (0.233), obligation-level Graph-RAG (0.663 [0.570–0.747]), and flat LLM judge / Claude Haiku (0.830 [0.747–0.908]). Three directional observations emerge at pilot scale ($n = 101$): SILENT is semantically detectable but invisible to entailment-only formulations; pair-level obligation graphs improve over lexical methods but trail flat-LLM context (CIs partially overlapping); and corpus-level graph construction is the indicated architectural target for large-scale silent-detection. RegDivergence-101 is a pilot release establishing the task formulation and baseline hierarchy; four unrepresented regulatory domains and an expansion roadmap are described in §8.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; • **Information systems** → *Document filtering*; • **Applied computing** → *Health informatics*.

Additional Key Words and Phrases: large language models, LLM benchmark, regulatory NLP, AI for compliance, FDA/EMA divergence, contradiction detection, Graph-RAG, retrieval-augmented generation, generative AI, life sciences, AGREE/DIVERGE/SILENT classification

1 INTRODUCTION

Bringing a medicine to market in both the US and EU requires satisfying two regulators that publish guidance separately and, frequently, *differently*. A concrete example: gene-therapy sponsors face a 15-year long-term follow-up

*Corresponding author.

Authors' Contact Information: Chuchu Wu, chuchuw@alumni.cmu.edu, Carnegie Mellon University, Pittsburgh, PA, USA; Zhiyin Zhou, zzhou20@pratt.edu, Pratt Institute, New York, NY, USA; Jingzhuo Hu, jingzhuoh00@gmail.com, University of Pennsylvania, Philadelphia, PA, USA; Liang You, liy121@pitt.edu, University of Pittsburgh, Pittsburgh, PA, USA.

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recommendation from FDA, while EMA defers the duration to case-by-case risk assessment—same topic, materially different specificity. Another documented divergence concerns ICH E9(R1) **estimand** implementation for oncology: although FDA, EMA, and PMDA jointly authored the ICH E9(R1) framework, *indication-specific implementation guidance* has diverged—for progression-free survival, EMA practice has defaulted to a treatment-policy strategy, whereas FDA cancer-endpoint guidance has applied censoring for intercurrent events such as subsequent anticancer therapy [18]. Other documented divergences span pediatric extrapolation (ICH E11A), endpoint definitions in ulcerative colitis, and the regulatory pathway for companion diagnostics. A sponsor who misses one of these divergences during protocol design can lose a year or more re-running a study.

Reconciling the two corpora is currently a manual, expert task. Yet the NLP building blocks for automating it all exist—document-level NLI (ContractNLI [9], DocNLI [21]), legal contradiction detection (LegalWiz [12], CLAUSE [15]), and Graph-RAG (Microsoft GraphRAG [5], GraphCompliance [4], RAGulating Compliance [1]). **No prior work combines them to detect divergence between two parallel regulatory corpora from different jurisdictions.** All existing contradiction work is *intra-document* (within one contract) or *single-jurisdiction* (contract-vs-statute).

This paper makes three contributions:

- (1) **Task formulation.** Cross-jurisdiction regulatory divergence detection as a three-way classification (AGREE / DIVERGE / SILENT) over pre-aligned FDA/EMA requirement pairs, grounded in published expert comparisons. The task is explicitly scoped to the *classification* stage given pre-aligned pairs; topic alignment is a non-trivial upstream problem addressed in §3.1 and §3.2. SILENT is treated as directional (SILENT_FDA vs. SILENT_EMA); we report per-direction F1 in Table 4.
- (2) **Pilot benchmark. RegDivergence-101**, a pilot 101-pair benchmark (59 from peer-reviewed FDA/EMA comparison studies, 42 derived from primary FDA/EMA/ICH guidance) spanning 13 topic areas, with a documented two-track pair-construction protocol, inter-annotator agreement ($\kappa = 0.85$), and bootstrap confidence intervals. Four regulatory domains are currently unrepresented; §8 describes the expansion roadmap.
- (3) **Baseline hierarchy with three directional observations.** Four methods systematically compared at pilot scale ($n = 101$), yielding directional empirical observations on the task’s structure and the path to a corpus-level solution. Observations are reported with bootstrap 95% CIs; confirmatory significance testing awaits a larger benchmark.

2 RELATED WORK

Document-level NLI. ContractNLI [9] introduced document-level entailment/contradiction/not-mentioned classification with evidence extraction, and showed that contradiction detection lags entailment, especially under “negation by exception.” DocNLI [21] extended NLI to full documents.

Legal contradiction detection (2025–2026). LegalWiz [12] uses a multi-agent generation framework for legal contradiction detection but is confined to single-document regimes. Better Call CLAUSE [15] benchmarks 7,500+ perturbed contracts across ten anomaly categories and finds that LLMs miss subtle errors and struggle to *justify* them legally. Contradiction Detection in RAG Systems [6] uses LLMs as context validators. Two recent RAG-faithfulness studies bear directly on our LLM-judge and graph-RAG results: Chen et al. [3] probe whether a RAG system can recognise when retrieved context is incorrect, and Qian et al. [14] show that the surface relevance of retrieved evidence does not guarantee it *warrants* a claim—the same failure mode documented in our Observation 2, where modal-register similarity (RECOMMENDED vs. MANDATORY) misleads the pair-level graph.

Graph-RAG and regulatory compliance. Microsoft GraphRAG [5] builds entity graphs with community summaries for local and global queries. GraphCompliance [4] aligns *policy graphs* (regulatory requirements) with *context graphs* (organisational facts) for GDPR; it explicitly lists “handling conflicting regulatory interpretations” as open. RAGulating Compliance [1] builds an ontology-free regulatory knowledge graph for traceable QA.

Cross-jurisdiction regulatory benchmarks. The closest concurrent work is **Sino-US-DrugQA** [19], a bilingual (EN/ZH) benchmark of 11,871 multiple-choice QA items contrasting US (CFR Title 21) and Chinese (NMPA) drug regulation, with explicit *comparative* (4,310 items) and *parallel* subsets; the authors report that cross-jurisdictional comparative reasoning is measurably harder than monolingual retrieval. It shares our core thesis—concept-level alignment across non-isomorphic regulators—but differs in jurisdiction pair (US–CN vs. US–EU), framing (MCQA vs. three-way AGREE/DIVERGE/SILENT), and crucially cannot express the SILENT class, which is central to our formulation. Beyond NLP benchmarks, a substantial comparative-regulatory-science literature documents requirement divergence across many jurisdiction pairs—FDA/EMA/PMDA dosing and oncology labelling [7, 13], Health Canada vs. FDA/EMA submissions [11], and multi-country product-information corpora [16]—but these are human-authored comparisons, not machine-learning benchmarks. RegDivergence-101 is, to our knowledge, the first benchmark to operationalise FDA/EMA divergence as a three-way classification task with an explicit absence (SILENT) class.

Gap. Contradiction detection is intra-document or single-jurisdiction; Graph-RAG targets single-corpus QA or single-jurisdiction compliance; FDA/EMA divergence is documented manually but never automated. Our task sits precisely in this gap.

3 TASK AND DATASET

3.1 Task Definition

For a topic t , let f_t be an FDA requirement statement and e_t the corresponding EMA statement. The label y_t is:

- **AGREE** — both jurisdictions require substantively the same thing (same modal strength, same threshold, same scope of applicability). Paraphrase equivalence and synonymous modals (*shall* / *must*) are treated as AGREE; minor drafting differences that do not alter sponsor obligations are disregarded.
- **DIVERGE** — the requirements directly conflict: they impose different thresholds, different modal strengths (*shall* vs. *should*), different timelines, or mutually incompatible design specifications.
- **SILENT** — one jurisdiction does not address what the other regulates. SILENT is *directional* by construction:
 - **SILENT_FDA**: FDA is silent on this topic; an EMA requirement exists but no corresponding FDA requirement was found in the full published FDA corpus (guidance documents, Q&As, MaPPs).
 - **SILENT_EMA**: EMA is silent on this topic; an FDA requirement exists but no corresponding EMA requirement was found in the full published EMA corpus (guidelines, Q&As, reflection papers).

RegDivergence-101 records direction in the `silent_direction` field and reports per-direction F1 in Table 4. A unified **SILENT** label ($= \text{SILENT_FDA} \cup \text{SILENT_EMA}$) is retained in Table 3 for backward compatibility.

Epistemic status of SILENT labels. A SILENT label is an absence claim: it asserts that no counterpart requirement was found after an exhaustive corpus search. This carries irreducible epistemic uncertainty. We bound this uncertainty through (a) a documented corpus-search protocol (§3.3) and (b) a per-pair `doc_confidence` flag (HIGH / MODERATE). Downstream work should treat MODERATE-confidence SILENT pairs as soft labels.

Task scope and assumptions. This formulation assumes requirement pairs are *pre-aligned*—both the FDA and EMA sides address the same regulatory topic. Topic alignment is a non-trivial upstream problem: FDA and EMA

guidance frequently address overlapping domains at inconsistent levels of specificity, making it unclear whether two requirements are “on the same topic” at all. RegDivergence-101 is constructed from expert-panel comparisons that *already provide aligned pairs* and from a two-track alignment procedure described in §3.2; we treat topic alignment as a separate, upstream task and refer readers to §8 for the corpus-level architecture that would encompass it.

3.2 Pair Construction Protocol

RegDivergence-101 is constructed via two tracks that differ in how topical correspondence between FDA and EMA requirements is established. Per-pair source-track attribution (track: A or B) is released with the dataset.

Track A — expert-comparison-derived pairs (59 pairs, 58.4%). All 59 pairs were extracted from three open-access peer-reviewed studies whose explicit purpose was comparative mapping of FDA and EMA requirements in a defined therapeutic domain:

- (1) Vieujean et al. [20] — ulcerative colitis endpoints and trial design (40 pairs).
- (2) Boesen et al. [2] — psychiatric drug-trial guidelines (10 pairs).
- (3) Schwarz and Decristoforo [17] — radiopharmaceutical nonclinical requirements (9 pairs).

In each source paper, domain experts had already established co-topicality through structured literature mapping or side-by-side dossier comparison. Our annotation task for Track A was to classify the *relationship* (AGREE / DIVERGE / SILENT) between requirements that the source authors had already deemed co-topical. We inherit, document, and cite their alignment judgements; per-pair source-study attribution is released with the dataset.

Track B — primary guidance pairs (42 pairs, 41.6%). For pairs derived from ICH E9(R1), E11A [8], and FDA/EMA biosimilar and gene-therapy guidance, topical alignment was established in two steps:

- (1) *Anchor identification.* Requirements were associated with ICH Common Technical Document (CTD) module and sub-topic designations (e.g., Module 5.3.5.1 for comparative bioavailability study reports). Requirements sharing the same CTD anchor were candidate pairs.
- (2) *Expert confirmation.* The primary annotator independently reviewed each candidate pair and confirmed or rejected co-topicality before the pair entered the dataset.

Specificity exclusion criterion. A candidate Track B pair was excluded if the FDA and EMA requirements operated at different levels of the regulatory hierarchy (e.g., one requirement addressing trial design at the *statistical analysis plan* level while the other addressed the same concept at the level of *general statistical principles*). Such pairs would conflate regulatory level-of-detail differences with substantive DIVERGE labels. A small number of candidate pairs were excluded on this basis (the count and representative cases are documented in the released dataset’s `excluded_pairs.jsonl` file). The complete exclusion-criteria checklist is released with the dataset so third parties can audit or extend our alignment judgements.

Each pair records {id, topic, track, source, fda_text, ema_text, label, silent_direction, doc_confidence, disputed, annotation_notes}. The combined release contains **101 pairs**: 42 AGREE, 38 DIVERGE, 21 SILENT, across 13 topic areas. We report macro-F1 to weight all classes equally.

3.3 Annotation Protocol

Annotator background. Both annotators are industry practitioners whose work centres on regulatory and clinical-trial documentation. Annotator A (C. Wu, corresponding author) is a solution architect for a regulatory-document AI platform with sustained professional exposure to FDA/EMA guidance and submission workflows. Annotator B

(Z. Zhou) is a product manager for an AI-powered clinical-trial management platform spanning regulatory and eSource workflows. Neither holds formal regulatory-affairs officer credentialing; the peer-reviewed expert-panel source papers (Track A) provide the external human-expert anchor for the seed labels, and both annotators worked with the full relevant guidance document text, associated Q&As, and ICH technical documentation for each pair. Wu’s judgement was exercised principally on Track B pairs and on borderline Track A cases where source language was ambiguous.

Annotation stages. Labels were assigned in two stages.

Stage 1 (primary labels): Wu extracted requirement pairs from Track A source papers and Track B primary guidance. Labels for Track A were directly inferred from the expert authors’ stated assessments; Track B labels were assigned against the full guidance text using a definitions-based framing (“*do FDA and EMA impose the same substantive requirement?*”).

Stage 2 (IAA check): Zhou re-labelled all 101 pairs independently using a sponsor-obligation framing (“*would a global sponsor face different regulatory obligations when filing simultaneously with FDA and EMA?*”). The two framing perspectives are deliberately different: Stage 1 asks about regulatory equivalence; Stage 2 asks about operational impact for sponsors. This design probes whether the κ reflects a stable, framing-independent signal rather than an artefact of one framing choice.

IAA design, scope, and interpretation. The two annotation passes use deliberately different **framing perspectives**—definitions-based (WU) vs. sponsor-obligation (ZHOU)—to probe whether the label signal is stable across framing choices rather than an artefact of one framing. This design follows the two-framing paradigm of NLI benchmarks such as MedNLI and ContractNLI. Two interpretive caveats must be stated transparently:

- **Co-author annotators.** Both WU and ZHOU are co-authors of this paper. IAA between co-authors may exhibit conformity bias absent from independent annotation. We report the coefficient honestly but recommend readers treat it as a lower-bound consistency estimate; an independent third-annotator validation pass is planned for the RegDivergence-500 release.
- **Cross-framing coefficient.** Because the two stages use different framing prompts, $\kappa = 0.85$ measures agreement *across* two cognitive framings of the same task, not agreement between two annotators performing an identical task. Standard IAA assumes identical instructions; our coefficient is best interpreted as a cross-framing consistency measure. The two framings can in principle produce different labels on the same pair (a requirement that is substantively equivalent might still create different filing obligations), so this framing gap is a feature of the design, not an oversight.

Disagreements were resolved through a separate adjudication step (§?? below).

Annotation history. The R1 submission reported a secondary prompt-robustness check: single-LLM re-annotation of a 36-pair stratified sample under a deliberately different prompt, yielding $\kappa = 0.542$ (moderate; test-retest reliability for the LLM baselines, not a human labelling-agreement measure). The R2 revision adds full human annotation by ZHOU over all 101 pairs (Stage 2 above). The $\kappa = 0.85$ figure is the human cross-framing consistency coefficient from this addition; the $\kappa = 0.542$ figure is retained as a separate LLM prompt-robustness statistic and is reported as such below. Both figures are real; they measure different things.

IAA results. Cross-framing consistency between WU and ZHOU over all 101 core pairs: observed agreement 0.90, Cohen’s $\kappa = 0.85$ (3-class; 0.86 on the directional 4-class scheme, consistent with high SILENT agreement inflating 4-class κ marginally). Per Landis & Koch (1977) [10]: *substantial*. AGREE and DIVERGE show the most variability across

framings ($\kappa \approx 0.79\text{--}0.81$); SILENT is near-perfectly agreed ($\kappa = 0.98$), reflecting the relative clarity of absence labels across framing perspectives.

Table 1 contextualises the dual-annotation $\kappa = 0.85$ against related annotation studies:

Table 1. IAA comparison across related NLP annotation studies.

Task	κ	Reference
Clinical NLI (MedNLI)	0.65	Romanov & Shivade (2018)
Legal NLI (ContractNLI)	~ 0.72	Koreeda and Manning
Biomedical contradiction	$\sim 0.55\text{--}0.60$	BioASQ organisers
Regulatory divergence (ours)	0.85	This work

Our dual-annotation $\kappa = 0.85$ is on par with or above medical (MedNLI) and legal (ContractNLI) NLI benchmarks, indicating that—once the label schema is fixed—FDA/EMA divergence can be annotated reliably. Per Landis and Koch [10], this falls in the *substantial* band (κ 0.61–0.80 and above). The lower prompt-robustness figure ($\kappa = 0.542$, single-LLM re-annotation) is reported separately above as a test-retest measure for the LLM baselines, not as the benchmark’s labelling agreement.

Disagreement resolution. Of the 101 pairs independently annotated by both WU and ZHOU, 10 showed disagreement (observed agreement 0.90).

- $n = 6$: source expert-panel language unambiguously supported one label; that label was adopted.
- $n = 4$: genuinely borderline; Stage 1 (WU) label retained; these pairs are flagged as `disputed = True` in the released dataset.

Written justifications for all 10 cases are released as `annotation_notes` per pair, enabling downstream users to distinguish clear-cut from contested cases.

Documentation-confidence protocol for SILENT pairs. For all 21 SILENT pairs, the annotator documented the corpus-search scope: whether the complete FDA corpus (guidance documents, Q&As, MaPPs) and complete EMA corpus (guidelines, Q&As, reflection papers) were searched before the silence conclusion was reached. Each SILENT pair carries a `doc_confidence` flag with two values:

- **HIGH**: complete relevant corpus searched; no counterpart found.
- **MODERATE**: corpus searched to best of annotator’s knowledge; non-zero uncertainty remains (e.g., recent unpublished positions, informal agency communications, draft guidance not yet finalised).

In the current pilot release, all 21 SILENT pairs are conservatively assigned MODERATE confidence: the FDA and EMA corpora are large and evolving, and absence claims cannot be made with absolute certainty from the published corpus alone. The HIGH/MODERATE distinction will be operationalised with per-pair search-documentation records in the RegDivergence-500 release. Downstream evaluations should stratify SILENT performance by `doc_confidence` once HIGH-confidence pairs are available.

4 BASELINES

4.1 Evaluation Context

RegDivergence-101 is an **evaluation benchmark**, not a training dataset. All four baselines are zero-shot or few-shot: the LLM judge uses no labelled pairs during inference; the NLI cross-encoder is DistilBERT-MNLI fine-tuned

on MultiNLI, not on our data; the Graph-RAG and lexical baselines require no supervised training on our pairs. The relevant comparator is therefore specialised NLI evaluation benchmarks, not large training corpora:

Table 2. RegDivergence-101 in context of other specialised NLI evaluation benchmarks.

Benchmark	Task domain	Dev/Test pairs
MedNLI	Clinical NLI	1,422
WinoGrande	Commonsense reasoning	1,267
CommonsenseQA	Commonsense QA	1,221
ContractNLI	Legal NLI	~1,100
BioASQ (contradiction)	Biomedical NLI	~600
RegDivergence-101	Regulatory divergence	101

Statistical adequacy. With $n = 101$ and a 1,000-resample bootstrap, macro-F1 95% CI half-widths are $\approx \pm 0.08$ – 0.10 . Adjacent methods in Table 3 are separated by $\Delta = 0.15$ – 0.60 macro-F1 (minimum separation: Graph-RAG vs. LLM judge, $\Delta = 0.167$, CIs partially overlapping), sufficient to establish the ranking order reported in §5 as a trend though not pairwise significance. We report these as a consistent ranking trend rather than a claim of statistical significance; a paired significance test on shared held-out predictions is left to future work.

4.2 Shared Evaluation Protocol

We evaluate four methods under a shared reporting framework: **stratified 5-fold splits** averaged over 5 random seeds, with **bootstrap 95% CI** (1,000 resamples). The protocol differs by method class. The lexical heuristic and NLI cross-encoder have tunable thresholds and use **nested** CV (inner folds tune thresholds, outer folds report). The Graph-RAG classifier and LLM judge have no tunable hyperparameters: the Graph-RAG classifier is reported over the same outer folds, and the LLM judge is scored directly on all 101 pairs. CIs for both LLM methods are bootstrap-only.

4.3 Lexical Heuristic

No neural model, no API—fully offline and deterministic. Combines TF-IDF cosine similarity with five conflict signals: negation asymmetry, divergence cue pairs (e.g., *single-arm* vs. *randomised*), modal-strength asymmetry (*must/shall* vs. *may/encouraged*), numeric-threshold mismatch, and one-sided specificity (concrete number vs. case-by-case deferral). A shared ICH citation is a strong AGREE signal.

4.4 NLI Cross-Encoder

typeform/distilbert-base-uncased-mnli, run in both directions; labels derived from contradiction/neutral/entailment probabilities with CV-tuned thresholds. We use DistilBERT-MNLI as a representative off-the-shelf NLI system. The structural SILENT gap observed in §7 (Observation 1) is model-agnostic: any NLI model lacking an explicit SILENT output class will map absence-of-regulation to *neutral*, collapsing SILENT recall. Substituting DeBERTa-v3-large or a modern cross-encoder would improve AGREE/DIVERGE F1 but would not resolve the SILENT structural gap without task-specific supervision.

4.5 Graph-RAG Pair-Level Classifier

For each text, Claude Haiku extracts a policy triplet $\langle \text{subject}, \text{obligation_level}, \text{requirement}, \text{conditions} \rangle$ where $\text{obligation_level} \in \{\text{MANDATORY}, \text{PROHIBITED}, \text{RECOMMENDED}, \text{PERMITTED}, \text{SILENT}\}$. Only MANDATORY $\leftrightarrow$ PROHIBITED is a

Table 3. Macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for the four baselines ($n = 101$ pairs).

Method	Macro-F1 [95% CI]	Acc	AGREE F1 [CI]	DIVERGE F1 [CI]	SILENT F1 [CI]
Lexical heuristic ¹	0.511 [0.411–0.605]	0.541	0.718 [0.605–0.809]	0.466 [0.320–0.593]	0.334 [0.133–0.537]
NLI cross-encoder ²	0.233 [0.152–0.303]	0.238	0.279 [0.103–0.449]	0.383 [0.247–0.506]	0.025 [0.000–0.086]
Graph-RAG pair-level ³	0.663 [0.570–0.747]	0.673	0.754 [0.646–0.851]	0.586 [0.433–0.722]	0.640 [0.478–0.771]
LLM judge (Haiku)⁴	0.830 [0.747–0.908]	0.842	0.905 [0.831–0.965]	0.805 [0.696–0.892]	0.778 [0.609–0.914]

¹Nested 5-fold CV, 5 seeds; thresholds tuned on inner folds only. ²Same CV protocol, thresholds tuned on inner folds. ³No tunable hyperparameters; bootstrap CI only. ⁴claude-haiku-4-5-20251001, default temperature; bootstrap CI only. Estimated cost: ~\$0.10 for all 101 pairs.

hard-coded DIVERGE rule; all other pairs are escalated to a second LLM call that reasons over the *obligation structure* of the aligned node pair, with explicit calibration that modal-register differences (RECOMMENDED vs. MANDATORY) do not automatically imply DIVERGE.

Evaluation note. The graph classifier was evaluated on the same 101 pairs as all other methods, using the same stratified nested CV outer folds for reporting. No hyperparameter tuning was performed (the MANDATORY ↔ PROHIBITED rule and the LLM prompt were fixed before evaluation began). Bootstrap CI is over the same held-out fold predictions as the other methods.

Within-model scope. Both the Graph-RAG classifier and the flat LLM judge use claude-haiku-4-5-20251001. The comparison between them is therefore a *within-model ablation*—with vs. without explicit obligation-graph structure—not an architectural comparison across model families. Conclusions about the relative value of graph structure should be confirmed with a second model family before generalisation.

4.6 LLM Judge (Flat Pairwise)

Claude Haiku with explicit AGREE / DIVERGE / SILENT definitions, prompted once per pair with both texts. No graph, no retrieval. Full prompt in Appendix A.

5 RESULTS

Table 3 reports macro-F1, accuracy, and per-class F1 with bootstrap 95% confidence intervals for all four methods.

Table 4. Per-direction SILENT F1 for all four methods (bootstrap 95% CI, $n = 101$ pairs). SILENT_FDA ($n = 13$): FDA is silent, EMA regulates. SILENT_EMA ($n = 8$): EMA is silent, FDA regulates.

Method	SILENT_FDA F1 [95% CI]	SILENT_EMA F1 [95% CI]
Lexical heuristic	0.277 [0.000–0.545]	0.393 [0.000–0.706]
NLI cross-encoder	0.049 [0.000–0.167]	0.000 [0.000–0.000]
Graph-RAG pair-level	0.716 [0.516–0.882]	0.535 [0.267–0.762]
LLM judge (Haiku)	0.795 [0.588–0.952]	0.738 [0.429–0.952]

SILENT-FDA: 13 pairs (FDA silent, EMA active). SILENT-EMA: 8 pairs (EMA silent, FDA active). Wide CIs on SILENT-EMA reflect the small stratum ($n = 8$). Bootstrap 1,000 resamples; stratified by 3-class label.

Point estimates increase monotonically from lexical to Graph-RAG to the LLM judge. At $n = 101$ the marginal bootstrap CIs partially overlap, so we report these as a consistent ranking trend rather than a claim of statistical significance; a paired significance test on shared held-out predictions is left to future work.

6 ERROR ANALYSIS: THE LEXICAL CEILING

The lexical heuristic misclassifies roughly 46 of the 101 pairs (accuracy 0.541). The illustrative errors below share a signature: **high lexical overlap, opposed regulatory stance**.

- **uc-sev-03 (DIVERGE → missed)**: FDA “modified Mayo score of 5 to 9” vs. EMA “full Mayo score of 9 to 12”—near-identical vocabulary, different scoring instrument and threshold.
- **uc-01**: Single-arm Mayo-score endpoint vs. co-primary modified-Mayo—same terms, different structural requirement.
- **gene-01**: FDA recommends 15 years; EMA defers to case-by-case—same topic, different specificity.

A bag-of-words model cannot resolve these; they require representing *what each requirement asserts about which entity*.

7 ANALYSIS

7.1 What the Baseline Hierarchy Reveals

Observation 1 — SILENT is semantically detectable but invisible to NLI framing. SILENT F1 = 0.025 under NLI (no native SILENT class) vs. 0.778 under the LLM judge (explicit definition). This contrast is large and consistent across bootstrap resamples; it reflects a structural constraint—the SILENT class requires a formulation that can express absence, which NLI entailment cannot—rather than a model-capacity effect. We note this observation is based on a distilbert-class model; a higher-capacity NLI encoder would improve AGREE/DIVERGE F1 but would not resolve the absence-detection gap without task-specific supervision (§7).

Observation 2 — For Claude Haiku on this dataset, pair-level obligation graphs improve over lexical (+0.15 F1) but trail flat prompting (−0.17 F1); CIs partially overlap at pilot scale. The graph makes the obligation structure explicit (MANDATORY / RECOMMENDED / etc.), which helps distinguish clear stance differences from similar-sounding text. But it discards full-text context that flat prompting uses to separate *same-requirement* / *different-modal-register* (AGREE) from *same-topic* / *different-threshold* (DIVERGE). Pairs like gene-01 (FDA 15 years vs. EMA case-by-case) both extract as RECOMMENDED and the graph calls them AGREE; the flat LLM correctly identifies the threshold conflict from the text [14]. *Scope note.* Both the graph classifier and the flat LLM judge use Claude Haiku; this comparison is a within-model ablation (with vs. without explicit graph structure) rather than a general claim about graph methods vs. LLM methods. Replication with a second model family is left to future work.

Observation 3 — Two distinct problem scopes suggest two distinct architectural targets. The *pair-level scope* — given pre-aligned sentences, label AGREE/DIVERGE/SILENT — is largely addressed by the flat LLM judge (0.830), approaching a task ceiling. The *corpus-level scope* — “is EMA silent on this topic *anywhere* across its guideline corpus?” — requires finding aligned pairs, not just classifying them; a pairwise LLM cannot address this, but a corpus-level policy graph can. These are different tasks. This paper establishes baselines for the pair-level scope; §8 motivates the corpus-level architecture as the next contribution. This observation is architectural and directional; it is not a validated finding.

7.2 Source-Stratified Performance Analysis

To assess whether performance reflects source-study-specific patterns rather than genuine task competence, we analyse performance stratified by track and by source concentration.

Track-level analysis. The LLM judge achieves:

- **Track A** (59 expert-comparison pairs): AGREE F1 = 0.769, DIVERGE F1 = 0.745.
- **Track B** (42 primary guidance pairs): AGREE F1 = 0.966, DIVERGE F1 = 0.923.

On the contamination-relevant AGREE/DIVERGE classes, performance is *not* higher on Track A (the higher source-study exposure split)—it is lower. The overall macro-F1 gap (0.630 Track B vs. 0.765 Track A) is explained by SILENT support ($n = 0$ in Track B vs. $n = 21$ in Track A), not by memorisation.

Source-concentration analysis. 40 of the 101 pairs originate from a single source document (Vieujean et al. UC endpoint paper). To test whether headline performance is driven by that stratum, we stratify the LLM judge:

- **JCC-2025 (UC paper, $n = 40$):** macro-F1 = 0.804; AGREE F1 = 0.842 ($n = 11$), DIVERGE F1 = 0.800 ($n = 14$), SILENT F1 = 0.769 ($n = 15$).
- **All other pairs ($n = 61$, Track A non-UC + all Track B):** macro-F1 = 0.844; AGREE F1 = 0.923 ($n = 31$), DIVERGE F1 = 0.810 ($n = 24$), SILENT F1 = 0.800 ($n = 6$).

Performance on the 61 non-JCC pairs (0.844) is marginally *higher* than on the concentrated JCC stratum (0.804), ruling out JCC-domain inflation as an explanation for the 0.830 headline. The slight gap is consistent with DIVERGE being harder in the UC-endpoint domain (many same-vocabulary, different-instrument pairs) than in primary guidance text.

Future work should evaluate on guidance issued *after* the model’s knowledge cutoff to probe temporal contamination independently.

8 FUTURE WORK

8.1 Corpus-Level Policy Graph

The pair-level experiments establish that the correct target for a genuine Graph-RAG contribution is **corpus-level** silent detection and divergence discovery—not re-ranking an already-aligned pair, but *finding* which FDA requirements have no EMA counterpart and vice versa. This architecture also subsumes the topic-alignment problem flagged in §3.1: alignment becomes a graph-construction step, not a pre-processing assumption.

The proposed architecture:

- (1) **Per-jurisdiction policy-graph construction**—index all FDA guidance documents and all EMA guidelines; extract requirement triplets at corpus scale.
- (2) **Cross-jurisdiction alignment**—entity normalisation + embedding similarity to link corresponding requirement nodes across the two graphs. This step directly addresses the pre-alignment assumption: alignment is performed by the system, not assumed.
- (3) **Divergence and silence discovery**—for each aligned node pair: DIVERGE if obligation structures conflict; SILENT if a concept node exists in one graph but has no aligned counterpart in the other.
- (4) **Explanation generation**—natural-language summaries for regulatory-affairs specialists.

Key open questions: (a) does a corpus-level graph recover SILENT cases that are undetectable pairwise? (b) what is the alignment precision when FDA and EMA use different terminology for the same concept? (c) can the graph be kept current as guidance is updated?

8.2 Benchmark Expansion

RegDivergence-101 is a **pilot release**. Four major regulatory domains are currently unrepresented: CMC (chemistry, manufacturing, and controls); clinical trial design beyond the covered indications (adaptive designs, estimand strategies, endpoint definitions); statistical analysis plan requirements (analysis populations, multiplicity, missing data); and labelling/SmPC requirements (indication wording, risk communication, package insert structure).

RegDivergence-500 is the planned v2.0 release: ≈ 500 pairs across all four unrepresented domains, seeded from additional expert-panel comparison studies and primary guidance text, with a community-annotation pipeline. Annotation guidelines are released alongside this paper to enable domain experts to contribute pairs from underrepresented areas. Human-human IAA is already reported at $\kappa = 0.85$ for the 101-pair core (§3.3); the v2.0 release will extend full dual-annotation to all expansion pairs and recruit independent regulatory-affairs professionals for a third-annotator validation pass.

Broadening the evidence base beyond three source studies. The pilot’s reliance on three FDA/EMA comparison studies is a scoping choice, not a ceiling on available evidence. We have identified a concrete sourcing roadmap that widens both the domain and the jurisdiction axes: (i) additional FDA/EMA expert comparisons across oncology, pediatrics, biosimilars, ATMPs, and labelling (a ~ 45 -source inventory compiled for this revision); (ii) a third jurisdiction (PMDA) via existing FDA/EMA/PMDA concordance studies on dosing and oncology labelling [7, 13], from which an ≈ 40 -pair FDA $\times$ PMDA set is extractable without changing the task definition; and (iii) raw label feeds—openFDA Structured Product Labeling and the open EMA SmPC corpus—paired by international nonproprietary name to mint candidate pairs at scale, and the MIT-licensed multi-country product-information corpus of Sartori et al. [16] as a secondary, instruction-level divergence benchmark. Drawing ground truth from many independent sources directly mitigates the risk that any single comparison study’s biases or omissions are embedded in the benchmark.

Initial expansion increment. As a concrete down-payment on this roadmap, we release **21 additional candidate pairs** spanning eight currently-unrepresented domains—pediatric development, biosimilars, advanced-therapy medicinal products, expedited approval pathways, nitrosamine impurity limits, dissolution statistics, quality-by-design, and companion diagnostics—each mined from a peer-reviewed FDA/EMA (or, for companion diagnostics, FDA/EU-IVDR) comparison study with an explicit side-by-side requirement contrast. Because comparison papers preferentially document differences, this increment is DIVERGE-heavy (18 DIVERGE, 3 AGREE; expert-adjudicated) and is released as *label-candidate* data pending dual-annotation by a second human annotator; it is held separate from the frozen 101-pair evaluation set rather than folded into it.

Cross-pair generalisation. Because the task framing (not the FDA/EMA pair specifically) is the contribution, a natural validation is to test whether our best system transfers to a different jurisdiction pair. We plan a zero-shot probe of the LLM judge on a size-matched *comparative* subset of Sino-US-DrugQA [19], reporting the cross-pair generalisation gap. We note two caveats: Sino-US-DrugQA is MCQA (accuracy, not macro-F1, and no SILENT class), so a symmetric comparison requires either restricting to a “which jurisdiction is stricter” subset or wrapping our items in an MCQA shell; and it is currently a preprint pending peer review.

9 LIMITATIONS

Topic alignment subjectivity. [R1] The “same topic” judgement for Track B pairs was made by one annotator using the CTD-anchor procedure documented in §3.2, and is not guaranteed to be fully reproducible without our exclusion-criteria checklist (released with the dataset). For Track A pairs, co-topicality is inherited from source expert studies

whose alignment methodology we do not control. Future work extending the benchmark should document alignment methodology explicitly and consider measuring inter-annotator agreement on the co-topicality judgement itself, separately from the AGREE/DIVERGE/SILENT classification.

SILENT epistemic uncertainty. [R2] SILENT labels carry irreducible epistemic uncertainty. We bound this through the exhaustive-search protocol (§3.3) and the `doc_confidence` flag, but MODERATE-confidence pairs remain epistemically soft and users should treat them accordingly. The corpus-level architecture (§8) is the principled long-term remedy: it replaces the absence claim from a pair-level judgement with systematic corpus indexing.

Benchmark scale and therapeutic-area coverage. [R3] 101 pairs from three source studies introduces domain concentration (40/101 from one UC comparison paper) and limits statistical power. Results should be treated as directional. The therapeutic areas covered—inflammatory bowel disease, psychiatry, radiopharmaceuticals, estimand methodology, pediatric extrapolation, biosimilars, gene therapy—exclude oncology, rare diseases, medical devices, and vaccines. RegDivergence-101 is a **pilot benchmark** for establishing the task formulation and baseline hierarchy; the RegDivergence-500 extension (§8.2) is required before deployment conclusions can be drawn across the full regulatory landscape.

Human-human IAA. [R4, resolved] This revision adds full human-human annotation by a second independent annotator (Z. Zhou), yielding Cohen’s $\kappa = 0.85$ (3-class; 0.86 directional) over all 101 pairs—*substantial* agreement per Landis and Koch [10]. The earlier $\kappa = 0.542$ (LLM test-retest on a 36-pair sample, §3.3) is retained in the paper as a prompt-robustness measure for the LLM baselines, not as the primary labelling IAA. Single-annotator limitations for the 21 expansion pairs remain; those will receive full dual-annotation in the RegDivergence-500 release.

SILENT operationalisation. [R2, existing] Our SILENT pairs use a placeholder on the absent side, which may not reflect real retrieval conditions where a system must determine absence from a full corpus. The pair-level SILENT task is a proxy; §8 describes the correct corpus-level formulation.

Data contamination. [existing] Claude Haiku may have been exposed to FDA/EMA guidance and the published expert-panel comparison papers during pretraining. The source-stratified analysis in §7.2 provides evidence against contamination-driven inflation of performance on Track A pairs. Future work should evaluate on guidance issued after the model’s knowledge cutoff.

Benchmark use norms. RegDivergence-101 is a **pilot benchmark**; three norms govern appropriate use:

- (1) *Insufficient for deployment decisions.* A score on RegDivergence-101 is not sufficient evidence of model fitness for any regulatory-affairs workflow. The benchmark measures pair-level classification given pre-aligned text; real-world workflows require alignment, retrieval, and error-consequence assessment.
- (2) *Not domain-generalizable without held-out evaluation.* The 40/101 domain concentration in the JCC-2025 UC paper means scores may not transfer to other therapeutic areas. Models should be evaluated on held-out domains before domain-general claims are made.
- (3) *Not a sole procurement criterion.* RegDivergence-101 should not be used as the sole criterion in AI system procurement for regulatory-affairs applications until the RegDivergence-500 extension provides broader coverage.

Single-model evaluation and LLM circularity. [new] All LLM-dependent components—graph triplet extraction, graph classification, flat LLM judge, and the secondary annotation-consistency check—use the same model family (Claude Haiku). The Graph-RAG vs. flat-LLM comparison is a within-model ablation. The secondary $\kappa = 0.542$ consistency check (LLM re-annotation, §3.3) may reflect the same model’s label tendencies rather than independent human-LLM agreement. Multi-model replication is planned for the RegDivergence-500 release.

Single jurisdiction pair. [R3] All results are on the FDA/EMA pair. Whether the baseline hierarchy and the SILENT-detection findings transfer to other jurisdiction pairs (e.g., FDA/PMDA, FDA/NMPA) is untested in this work. The cross-pair generalisation probe against Sino-US-DrugQA [19] described in §8.2 is the planned first test; until it is run, our conclusions should be read as specific to FDA/EMA.

10 CONCLUSION

We introduced cross-jurisdiction regulatory divergence detection, released **RegDivergence-101**, a 101-pair expert-grounded **pilot** evaluation benchmark ($\kappa = 0.85$, documented two-track pair-construction protocol, per-direction SILENT annotation), and characterised a four-method baseline hierarchy with bootstrap CIs. Three directional observations at pilot scale ($n = 101$, CIs partially overlapping for adjacent methods): SILENT requires explicit absence-aware formulation (NLI F1 0.03 $\rightarrow$ LLM F1 0.78); for Claude Haiku on this dataset, pair-level graph structure improves over lexical heuristics (+0.15 F1) but trails flat prompting (−0.17 F1); and the two-scope distinction (pair-level vs. corpus-level) motivates the corpus-level graph architecture as the next contribution. The flat LLM judge (0.830) establishes a pair-level performance reference at pilot scale; the corpus-level architecture proposed in §8 is the correct target for large-scale silence discovery. The benchmark is a pilot: four regulatory domains (CMC, clinical trial design, statistical analysis, labelling) remain unrepresented, and the planned RegDivergence-500 extension with expanded dual-annotation will establish the task at production scale.

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Acknowledgments This work was conducted independently. Benchmark pairs were derived from publicly available FDA/EMA guidance documents and open-access peer-reviewed comparison studies. LLM API calls used the Anthropic API (Claude Haiku, `claude-haiku-4-5-20251001`).

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A LLM JUDGE PROMPT

System:

You are a regulatory-affairs expert comparing FDA and EMA guidance documents. Given an FDA requirement and an EMA requirement on the same topic, classify their relationship.

Respond with ONLY a JSON object in this exact format:

```
{"label": "<AGREE|DIVERGE|SILENT>", "reason": "<one sentence>"}
```

Definitions:

- AGREE: both jurisdictions require substantively the same thing
- DIVERGE: the requirements directly conflict (one mandates what the other forbids or softens)
- SILENT: one jurisdiction does not address what the other regulates

User:

FDA requirement:

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```
{fda_text}
```

EMA requirement:

```
{ema_text}
```

Classify the relationship.

Model: claude-haiku-4-5-20251001. Temperature: default (1.0). Max tokens: 100. Results cached; each pair scored exactly once.

B GRAPH-RAG PROMPTS

The Graph-RAG classifier uses two sequential LLM calls per pair.

Stage 1 — Triplet Extraction (system):

You are a regulatory-affairs expert. Extract the policy structure from a regulatory requirement text.

Return ONLY a JSON object:

```
{
  "subject": "<who or what is regulated>",
  "obligation_level": "<MANDATORY|PROHIBITED|RECOMMENDED|
                      PERMITTED|SILENT>",
  "requirement": "<the core requirement in one clause>",
  "conditions": "<key qualifications, thresholds, or scope
                limits; empty string if none>"
}
```

obligation_level definitions:

- MANDATORY: must / shall / required / is required
- PROHIBITED: must not / prohibited / not permitted
- RECOMMENDED: should / encouraged / preferred / recommended
- PERMITTED: may / can / is acceptable / is allowed
- SILENT: the text does not address this topic

Stage 1 — User:

Extract policy structure from this [FDA/EMA] requirement:

```
{requirement_text}
```

Stage 2 — Graph Classification (system):

You are a regulatory-affairs expert comparing FDA and EMA policy graphs.

Given two extracted policy nodes (one FDA, one EMA) on the same topic, classify their relationship.

Return ONLY a JSON object:

```
{ "label": "<AGREE|DIVERGE|SILENT>",
  "reason": "<one sentence explaining the obligation-level
            comparison>" }
```

Definitions:

- AGREE: the requirements are substantively compatible. Different modal registers (should vs shall, recommended vs required) do NOT automatically mean DIVERGE.
- DIVERGE: the requirements genuinely conflict -- one jurisdiction mandates what the other forbids, sets a different numeric threshold, or imposes a requirement the other explicitly rejects.
- SILENT: one jurisdiction does not address this topic at all (obligation_level is SILENT or text is absent), while the other has a real requirement.

Key calibration: focus on WHAT is required, not HOW it is worded. Modal-language differences alone (RECOMMENDED vs MANDATORY) should lead to AGREE if the substantive requirement is the same.

Stage 2 — User:

FDA policy node:

```
subject: {fda_subject}
obligation: {fda_obligation}
requirement: {fda_req}
conditions: {fda_cond}
```

EMA policy node:

```
subject: {ema_subject}
obligation: {ema_obligation}
requirement: {ema_req}
conditions: {ema_cond}
```

Classify the relationship between these two obligation nodes.

Hard rule (no LLM call): MANDATORY $\leftrightarrow$ PROHIBITED is classified DIVERGE without LLM escalation. Both SILENT $\rightarrow$ SILENT is also handled by rule. All other obligation pairs escalate to Stage 2. Model: claude-haiku-4-5-20251001. Max tokens: 300.

C SAMPLE EXCLUDED PAIRS — TRACK B SPECIFICITY CRITERION

The following representative examples illustrate the specificity-exclusion criterion applied to Track B candidate pairs (§3.2). Each pair was rejected because the FDA and EMA requirements addressed the same regulatory concept at different levels of the regulatory hierarchy.

Table 5. Sample rejected Track B candidate pairs.

ID	FDA requirement (summary)	EMA requirement (summary)	Rejection reason
X1	FDA requires a pre-specified primary estimand in the SAP, including intercurrent event handling strategy (ICH E9R1 implementation).	EMA endorses the ICH E9R1 estimand framework at the general principles level.	FDA operates at SAP specification level; EMA at framework-endorsement level. Different regulatory layers.
X2	FDA specifies pediatric study reporting timeline (12 months post-approval, per PREA).	EMA requires a Paediatric Investigation Plan (PIP) at application submission (EU Regulation 1901/2006).	FDA is post-approval reporting; EMA is pre-application planning. Different procedural stages, not comparable obligations.
X3	FDA guidance on biosimilar analytical comparability (specific assay-panel requirements).	EMA biosimilar guideline endorses analytical comparability at the principle level without specifying assay panels.	FDA at assay-specification level; EMA at principle level. Specificity mismatch precludes AGREE/DIVERGE classification.

The complete excluded-pair list with per-pair rationale is released in `data/excluded_pairs.csv`. The full exclusion-criteria checklist is released in `data/alignment_protocol.md`.